Hypothesis one sample t-tests

1 A researcher decides to test $H_0 : \mu = 23.5$ against $H_1 : \mu < 23.5$. The sample size of 45 is found to have a mean of 22.9 and a standard deviation of 1.2.

a Find the value of the test statistic, t, correct to two decimal places.

b Complete the statement:

$$p\text{-value} = P\left(T \leq \text{▯}\right), T \sim t_{\text{▯}-1}.$$

c Find the p-value for the sample, correct to four decimal places.

d What does the p-value for the sample indicate about the null hypothesis. Is H_0 likely to be rejected or not rejected?

2 A researcher decides to test $H_0 : \mu = 55$ against $H_1 : \mu > 55$. The sample size of 60 is found to have a mean of 56.2 and a standard deviation of 4.2.

a Find the value of the test statistic, t, correct to two decimal places.

b Complete the statement:

$$p\text{-value} = P\left(T \geq \text{▯}\right), T \sim t_{\text{▯}-1}.$$

c Find the p-value for the sample, correct to four decimal places.

d If the significance level is $\alpha = 5\%$, will H_0 be rejected or not rejected?

3 A researcher decides to test $H_0 : \mu = 16$ against $H_1 : \mu \neq 16$. The sample size of 30 is found to have a mean of 15.9 and a standard deviation of 1.36.

a Find the value of the test statistic, t, correct to two decimal places.

b Complete the statement:

$$p\text{-value} = P\left(T > \text{▯}\right) \text{ or } p\text{-value} = P\left(T < \text{▯}\right), T \sim t_{\text{▯}-1}.$$

c Find the p-value for the sample, correct to four decimal places.

d If the significance level is $\alpha = 10\%$, will H_0 be rejected or not rejected?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Based on population figures and other general information on the Australian population, on average, a family of four in Australia spends about $\$1135$ annually on dental expenditures. A regional dental association suspects that this amount is not correct and wants to test if this figure is accurate for their area of country. To test this, 20 families of 4 are randomly selected from the population in that area of the country and a log is kept of the family's dental expenditure for one year. The resulting data are given below. Assume that dental expenditure is normally distributed in the population.

$$1217,\ 1333,\ 1219,\ 1515,\ 799,\ 1121,\ 1187,\ 872,\ 932,\ 810,$$
$$689,\ 912,\ 933,\ 939,\ 964,\ 979,\ 1595,\ 989,\ 1002,\ 984$$

a Define the hypothesis set for this problem.

b Find the t-statistic for this two-tailed hypothesis problem, correct to two decimal places.

c Find the p-value for the sample, correct to four decimal places.

d If the significance level is $\alpha = 10\%$, does it appear that the regional dental association is correct in its claim? Explain your answer.

8 The contents of 25 cans of soft drink labelled as $300\ \text{ml}$ were measured and results were as follows:

$$301,\ 298,\ 289,\ 302,\ 299,\ 295,\ 300,\ 301,\ 297,\ 298,\ 300,\ 301,$$
$$302,\ 299,\ 295,\ 303,\ 297,\ 301,\ 300,\ 302,\ 302,\ 299,\ 298,\ 296,\ 302$$

Quality control wants to check that can contents are accurate within a 1% significance level. Assume that can contents data is normally distributed.

a Define the hypothesis set for this problem.

b Find the p-value for the sample, correct to two decimal places.

c State the sample mean, correct to two decimal places.

d Considering the significance level does it appear that there is a problem with the can contents for this brand? Explain your answer.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647